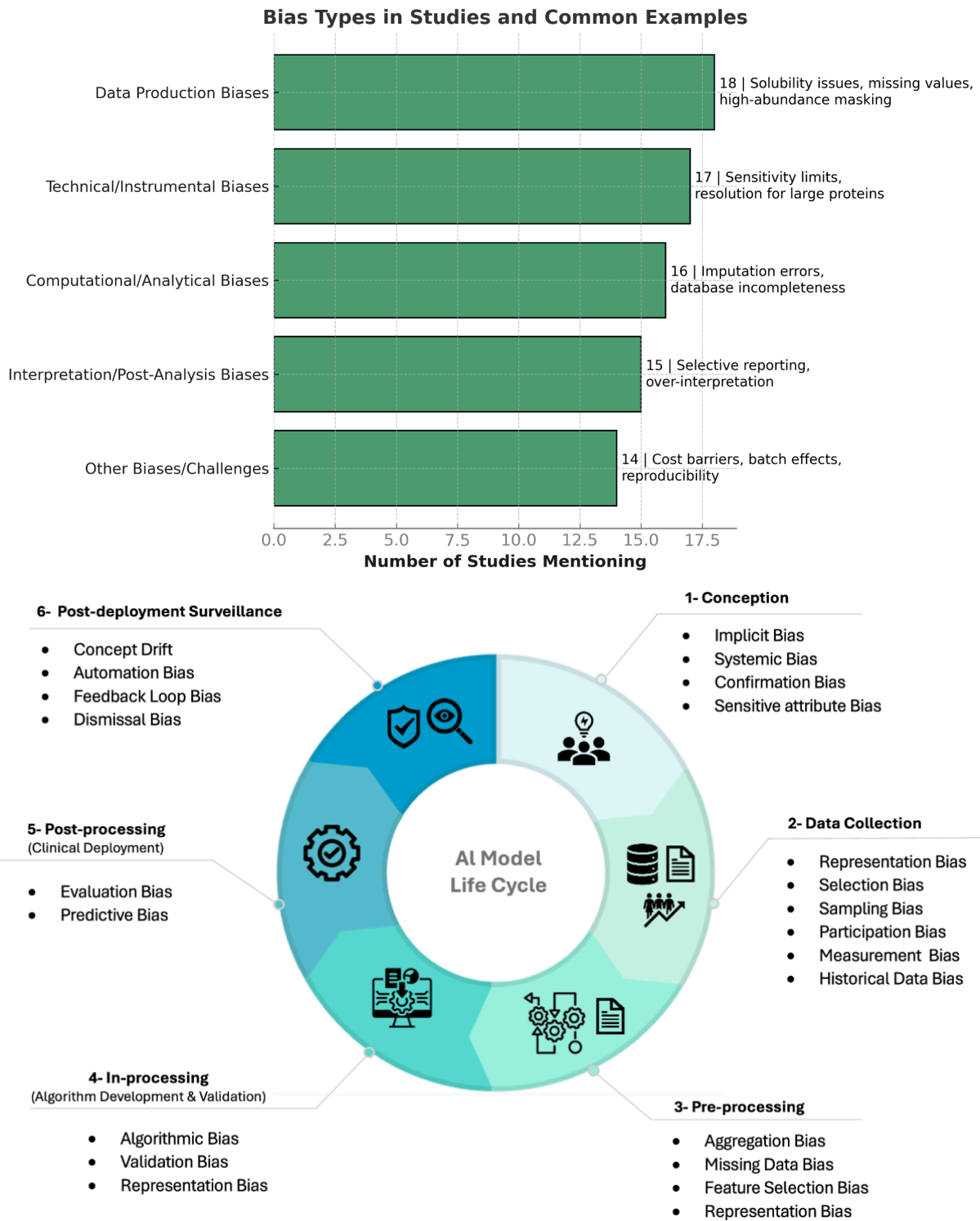




## Figure Set Overview Table

| Fig. ID | Title | Core Message | Supports Section |
|---------|-------|--------------|------------------|
|---------|-------|--------------|------------------|

|       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Fig.1 | PRISMA 2020 Flow                      | Shows systematic review screening (552 → 179 included)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Methods               |
| Fig.2 | Bias Lifecycle in Omics               | <p>Biases propagate across acquisition → preprocessing → modeling → AI</p> <p><b>Workflow Stages:</b> Biases can be mapped to different stages of the data lifecycle: data acquisition, processing, analysis, and interpretation.</p> <p><b>For Researchers (avoiding bias):</b> A detailed <b>Bias Lifecycle Process Map</b> or <b>Pipeline Visualization</b> would be most helpful. This visualization would illustrate where biases emerge throughout the data lifecycle, from initial sample collection to final data interpretation, allowing researchers to identify and proactively mitigate issues at each stage. Including specific examples of biases and their solutions would make it highly practical.</p> | Background/Objective  |
| Fig.3 | Tree of Omics Biases                  | Taxonomy beyond representativeness: Data, Technical, Analytical, Infrastructure, Cultural                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Results               |
| Fig.4 | Cross-Omics Bias Landscape            | Frequency/intensity of biases across genomics, transcriptomics, proteomics, metabolomics, microbiome, general omics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Results               |
| Fig.5 | English vs Chinese Literature         | Unique & overlapping biases, with TCM and linguistic issues highlighted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Results               |
| Fig.6 | Microbiome Taxonomy Biases            | Dual nomenclature (ICNP vs SeqCode) and uncultured species exclusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Results               |
| Fig.7 | AI Modeling Bias & Validation         | Calibration, fairness disparity, and schematic of batch effects/data leakage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Results               |
| Fig.8 | Bias → Mitigation → Guideline Roadmap | Connects bias classes to mitigation strategies and TRIPOD-AI-like guideline adoption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Discussion/Conclusion |

**Optional Extended Set:** Suppl.Fig.S1–S3 cover per-omics subgroup bias counts, Chinese metabolomics examples, and full bias extraction table.

- **Temporal Trends:** The publication year for each of the 179 articles can be used to show how certain biases or research areas have evolved over time.
- **Bias Relationships:** *The data can illustrate how different biases are interconnected or how one type of bias can lead to another. For example, pre-analytical variability can affect multiple Omics platforms differently, creating platform-specific biases. (possible?)*

## Main Figure Specs

**Fig.1. PRISMA 2020 Flow**

- **Purpose:** Document transparent screening path from 552 to 179 included.
  - **Narrative:** Classic PRISMA funnel with placeholders where intermediate counts are needed.
  - **Visual Form:** PRISMA 2020 template.
  - **Panel Plan:** Single schematic.
  - **Data/Inputs:** 552 identified → duplicates removed (~XX), screened (~XX), excluded (~XX), final 179.
  - **Encodings:** Boxes, arrows, counts.
  - **Accessibility:** High-contrast, sans serif labels.
  - **Caption Draft:** “PRISMA 2020 flow diagram showing screening of 552 records to 179 included articles.”
  - **Implementation:** Use PRISMA 2020 R package or Inkscape template.
  - **Risks/Alt:** Confusion if placeholders not marked; alternative = text-table in supplement.
-

## Fig.2. Bias Lifecycle in Omics

- **Purpose:** Show where and how bias emerges across omics workflow.
  - **Narrative:** Pipeline from sample recruitment → sequencing → preprocessing → AI model → clinical translation; overlay bias points.
  - **Visual Form:** Annotated pipeline/flowchart.
  - **Panel Plan:**
    - A) Workflow pipeline
    - B) Callouts showing bias propagation arrows.
  - **Inputs:** Bias categories aligned to lifecycle stage.
  - **Encodings:** Horizontal pipeline, icons per stage.
  - **Accessibility:** Colorblind palette; icons aid recognition.
  - **Caption Draft:** “Biases can be introduced at multiple stages of the omics lifecycle and propagate into AI models.”
  - **Implementation:** Mermaid flowchart.
  - **Risks/Alt:** Overcrowding; alternative = multi-panel breakdown.
- 

## Fig.3. Tree of Omics Biases

- **Purpose:** Establish taxonomy beyond representativeness.
- **Narrative:** Hierarchical tree with five trunks: Data Production, Technical, Analytical, Infrastructure, Cultural.
- **Visual Form:** Radial tree diagram.
- **Panel Plan:** Single schematic.
- **Inputs:** Bias categories/sub-biases (e.g., sequencing limits, calibration issues).
- **Encodings:** Root node = Omics Bias; branches with labeled sub-nodes.
- **Accessibility:** Thick lines, consistent branch angles.

- **Caption Draft:** “Hierarchical taxonomy of bias in omics research, spanning technical to cultural sources.”
  - **Implementation:** Mermaid `mindmap` diagram.
  - **Risks/Alt:** Tree too dense; alternative = table + collapsed tree.
- 

#### Fig.4. Cross-Omics Bias Landscape

- **Purpose:** Compare prominence of bias types across omics domains.
  - **Narrative:** Quantitative frequency/intensity overview.
  - **Visual Form:** Heatmap or radar chart.
  - **Panel Plan:**
    - A) Heatmap (bias categories × omics types)
    - B) Optional radar overlay.
  - **Inputs:** Counts from 179 articles (to be populated).
  - **Encodings:** Gradient color scale or radial axes.
  - **Accessibility:** Colorblind-safe sequential palette.
  - **Caption Draft:** “Distribution of bias types across omics domains reveals domain-specific hotspots.”
  - **Implementation:** ggplot2 heatmap.
  - **Risks/Alt:** Heatmap unreadable if sparse; alternative = stacked bars.
- 

#### Fig.5. English vs Chinese Literature

- **Purpose:** Highlight overlaps and unique bias themes.
- **Narrative:** Emphasizes TCM metabolomics gaps and reproducibility, absent in English.
- **Visual Form:** UpSet plot (preferred) or Venn diagram.

- **Panel Plan:**
    - A) Overlap schematic
    - B) Callouts: TCM database gaps, reproducibility issues.
  - **Inputs:** Bias lists coded by language source.
  - **Encodings:** Sets, intersection sizes.
  - **Accessibility:** Distinct shapes, not just colors.
  - **Caption Draft:** “Overlap and unique bias themes across English vs Chinese reviews, with TCM and reproducibility issues highlighted.”
  - **Implementation:** UpSetR.
  - **Risks/Alt:** Venn oversimplifies; keep UpSet for clarity.
- 

## Fig.6. Microbiome Taxonomy Biases

- **Purpose:** Show dual nomenclature and exclusion of uncultured species.
  - **Narrative:** Illustrates split between ICNP and SeqCode paths.
  - **Visual Form:** Sankey diagram with side callouts.
  - **Panel Plan:**
    - A) Sankey: ICNP vs SeqCode
    - B) Callout: uncultured species
    - C) Callout: linguistic barriers.
  - **Inputs:** Bias themes from microbiome studies.
  - **Encodings:** Flow thickness (frequency).
  - **Accessibility:** High-contrast flows, icons for bacteria.
  - **Caption Draft:** “Bias in microbiome taxonomy arises from dual nomenclature systems and exclusion of uncultured species.”
  - **Implementation:** Plotly Sankey.
  - **Risks/Alt:** Complexity; alternative = side-by-side schematics.
-

## Fig.7. AI Modeling Bias & Validation

- **Purpose:** Convey AI-specific risks (calibration, fairness, batch effects).
  - **Narrative:** Technical visualization of model bias.
  - **Visual Form:** Multi-panel: calibration curve, disparity bars, UMAP schematic.
  - **Panel Plan:**
    - A) Calibration curve (expected vs observed)
    - B) Fairness disparity bar plot
    - C) UMAP showing batch-aligned clusters.
  - **Inputs:** Synthetic placeholders or schematic.
  - **Encodings:** Axes: predicted vs observed, disparity bars.
  - **Accessibility:** Clear markers, colorblind palette.
  - **Caption Draft:** “AI models inherit bias via miscalibration, group disparities, and batch effects.”
  - **Implementation:** matplotlib plots + Illustrator schematic.
  - **Risks/Alt:** Readers may expect real data; clarify schematic/synthetic.
- 

## Fig.8. Bias → Mitigation → Guideline Roadmap

- **Purpose:** Link bias taxonomy to mitigation strategies and TRIPOD-AI-like path.
- **Narrative:** Forward-looking framework.
- **Visual Form:** Sankey/roadmap hybrid.
- **Panel Plan:**
  - A) Bias classes
  - B) Mitigation strategies
  - C) Future guideline adoption.
- **Inputs:** Derived mapping from discussion.
- **Encodings:** Flows with arrows; staged roadmap.
- **Accessibility:** Sequential shades, text callouts.

- **Caption Draft:** “Proposed roadmap linking bias classes to mitigation strategies and guideline adoption.”
- **Implementation:** Mermaid flow diagram.
- **Risks/Alt:** Overcrowding; alternative = table + schematic arrow.

## Supplementary Figures

- **Suppl.Fig.S1:** Per-omics subgroup article counts (bar chart).
- **Suppl.Fig.S2:** Case examples of TCM metabolomics reproducibility issues (schematic + photo icons).
- **Suppl.Fig.S3:** Extended bias extraction table with category frequency.
- **Suppl.Fig.S4 (optional):** Sensitivity analysis of inclusion/exclusion by language.

## Mini Style Guide

- Color palette: ColorBrewer “Set2” (colorblind-safe).
- Use consistent muted background, no heavy gridlines.
- Line weights: 1.2 pt main, 0.8 pt secondary.
- Font: Source Sans Pro, 8 pt minimum.
- Panel labels (A, B, C) in bold, top-left, 12 pt.
- Icons: sequencing machine, computer, globe, database.
- Heatmaps: sequential (light-dark), no rainbow.
- Radar/UpSet: use shape + texture to disambiguate.
- Callouts: gray connector lines with small arrowhead.



- Captions ≤60 words, descriptive not interpretive.
  - Vector formats: export to PDF/SVG.
  - Avoid overlapping labels; use rotated text for dense heatmaps.
- 

#### Opus 4.1

Based on your systematic review about bias in omics data, I recommend the following figures that would effectively communicate your findings:

#### ## Core Figures

##### **\*\*Figure 1: PRISMA Flow Diagram\*\***

Standard flowchart showing the systematic review process: 552 initial papers → duplicate removal → screening → final 179 articles included. Essential for any systematic review.

##### **\*\*Figure 2: Tree-Like Structure of Biases\*\***

Your proposed hierarchical visualization showing bias categories branching from main trunk to specific types. This would be your key conceptual contribution, organizing biases by:

- Data production level
- Technical/instrumental level
- Computational/analytical level
- Infrastructure/equity level

##### **\*\*Figure 3: Bias Distribution Heatmap\*\***

Matrix visualization with omics types (genomics, transcriptomics, proteomics, metabolomics) on one axis and bias categories on the other. Color intensity indicates prevalence or frequency of each bias type per omic domain.

##### **\*\*Figure 4: Geographic and Language Distribution Map\*\***

World map showing article distribution by country/region, with separate indicators for English vs Chinese literature. Could include pie charts showing bias types identified uniquely in Chinese vs English literature.

##### **\*\*Figure 5: Timeline of Bias Evolution\*\***

Temporal analysis (2019-2024) showing how attention to different bias types has changed over time, potentially revealing emerging concerns or persistent challenges.

## ## Supporting Figures

### **\*\*Figure 6: Mitigation Strategy Framework\*\***

Flowchart or circular diagram connecting specific biases to their corresponding mitigation approaches, showing the pathway from bias identification to solution implementation.

### **\*\*Figure 7: Comparison of Bias Patterns\*\***

Radar or spider plot comparing bias profiles across different omics types, making it easy to see which omics domains are most affected by specific bias categories.

### **\*\*Figure 8: AI Integration Points\*\***

Diagram showing where AI intersects with the omics pipeline and where biases can be amplified or mitigated at each stage.

These figures would provide both comprehensive coverage of your findings and clear visual communication of your novel contributions, particularly the bias taxonomy and the inclusion of Chinese literature perspectives.